

$$\begin{aligned}
54\bar{I}_1 - 50\bar{I}_2 - 3\bar{I}_3 &= 115\angle 0^\circ \quad - \langle i \rangle \\
-50\bar{I}_1 + (170 + j10)\bar{I}_2 - 100\bar{I}_3 &= 0 \quad - \langle ii \rangle \\
-3\bar{I}_1 - 100\bar{I}_2 + 104\bar{I}_3 &= 115\angle 0^\circ \quad - \langle iii \rangle
\end{aligned}$$

From (ii) -

$$\begin{aligned}
\bar{I}_1 &= \frac{100\bar{I}_3 - (170 + j10)\bar{I}_2}{-50} \\
&= -2\bar{I}_3 + \left(\frac{17}{5} + j\frac{1}{5}\right)\bar{I}_2 \quad - \langle iv \rangle \\
54 \left[-2\bar{I}_3 + \left(\frac{17}{5} + j\frac{1}{5}\right)\bar{I}_2 \right] - 50\bar{I}_2 - 3\bar{I}_3 &= 115\angle 0^\circ \\
\Rightarrow -111\bar{I}_3 + (133.6 + j10.8)\bar{I}_2 &= 115\angle 0^\circ - \langle v \rangle \\
-3 \left[-2\bar{I}_3 + \left(\frac{17}{5} + j\frac{1}{5}\right)\bar{I}_2 \right] - 100\bar{I}_2 + 104\bar{I}_3 &= 115\angle 0^\circ \\
\Rightarrow 110\bar{I}_3 - (110.2 + j0.6)\bar{I}_2 &= 115\angle 0^\circ - \langle vi \rangle
\end{aligned}$$

$\langle v \rangle + \langle vi \rangle$ we get -

$$\begin{aligned}
-\bar{I}_3 + (23.4 + j10.2)\bar{I}_2 &= 230\angle 0^\circ \\
\Rightarrow \bar{I}_3 &= (23.4 + j10.2)\bar{I}_2 - 230\angle 0^\circ \quad - \langle vi \rangle
\end{aligned}$$

Putting value of $\bar{I}_3$ in $\langle vi \rangle$ -

$$\begin{aligned}
[(23.4 + j10.2)\bar{I}_2 - 230\angle 0^\circ] 110 - (110.2 + j0.6)\bar{I}_2 &= 115\angle 0^\circ \\
\Rightarrow (2463.8 + j1121.4)\bar{I}_2 &= 25415\angle 0^\circ \\
\Rightarrow \bar{I}_2 &= 9.389\angle -24.47^\circ \text{ A.}
\end{aligned}$$